

INSTRUCTIONS

- All questions are multiple choice with exactly one correct option (A-D).
- Mark your answers on the provided OMR sheet only — do not write answers on this paper.
- Marks for each question are indicated against it. There is no negative marking unless announced.

1. Suppose you are building an Authentication Service, what will be the most secure way of storing passwords in the backend? [3 marks]

- A) Encrypt password with a global symmetric key
- B) Store password as binary data
- C) Encode password using base64
- D) Hash password with a unique salt for each password

2. In a 4-node(identical servers) system, after electing Node B as the leader, the system notices that Node B is slow to respond and hasn't made any decisions in a while. The other nodes, A, C, and D, continue to wait for Node B to act.

Which of the following might happen next? [3 marks]

- A) The system will continue waiting indefinitely for Node B to take action.
- B) The system will automatically demote Node B and re-elect a new leader after a timeout.
- C) Node A will take over leadership responsibilities without a formal election.
- D) The system will send more votes to Node B to encourage faster decision-making.

3. Would having a lot of unused data in a table affect query performance? [3 marks]

- A) No, queries fetch only the required data from disk
- B) Yes, Index and Sequential Scan can become inefficient
- C) Only sequential scans will be inefficient.
- D) Only Index scan will be inefficient.

4. An OS scheduler uses strict priority scheduling on a single core. If a low-priority process is ready to run, but a higher-priority process arrives while the low-priority process is running, what happens? [3 marks]

- A) The low-priority process continues until it finishes.
- B) The low-priority process is preempted, and the high-priority process runs.
- C) Both processes run simultaneously.
- D) The high-priority process is put on hold until the low-priority process finishes.

5. What part of ACID properties does a RAM based database sacrifice? [3 marks]

- A) atomicity
- B) consistency
- C) isolation
- D) durability

6. Consider the following example, when you have to run an executable on your machine which of the following is a must? [3 marks]

- A) The executable should have been built for the OS on your machine
- B) The executable should have been built for your processor architecture.
- C) The executable should have been built for both OS and processor architecture.
- D) None of the above matters, you can just run it

7. Consider a system running three processes (P1, P2, P3) with equal priority using round-robin scheduling with a time slice of 4 units. The processes have burst times of 8, 4, and 6 units respectively. What is the turnaround time for P3? [3 marks]

- A) 12 units
- B) 14 units
- C) 16 units
- D) 18 units

8. Two friends agree to meet at a café between 3:00 PM and 4:00 PM. Each arrives at a random time and waits for 10 minutes. What is the probability that both meet at the cafe? [5 marks]

- A) 0.31
- B) 0.33
- C) 0.28
- D) 0.25

9. Two players play a game where Player 1 rolls two fair 6-sided dice and Player 2 rolls fair 6-sided one die. Player 1 wins if the sum of their dice is greater than Player 2's die. What is the probability that Player 1 wins? [3 marks]

- A) 181/216
- B) 173/216
- C) 190/216
- D) 165/216

10. A tank has water filled in it and kept on a frictionless surface. How much time does it take to empty it? The Tank is cylindrical of height $h = 10$ m and diameter $d = 20$ cm. How much time will it take to empty the tank? There is a hole of diameter 2 cm. Density of liquid = 1 Kg/m^3 , $g = 10 \text{ m/s}^2$, Mass of tank = $\pi/100 \text{ kg}$ [7 marks]

- A) 141.4 sec
- B) 173.2 sec
- C) 200 sec
- D) 100 sec

11. A conical frictionless vessel with base facing up and tip facing down is set up on the ground. The radius at the top is $r_1 = 5$ m and radius at the base is $r_2 = 3$ m (imagine that the tip of the cone is cut). Height of the cut piece is $h = 4$ m. This vessel is rotating at angular velocity of $\Omega = 10 \text{ (s}^{-1}\text{)}$. A mass is kept at the slant surface from the top larger end of the cone. Time taken by the mass to reach the ground? $G = 10 \text{ m/s}^2$, $m_1 = 1 \text{ Kg}$ [3 marks]

- A) $2/\sqrt{5}$ sec
- B) 1 sec
- C) Will not reach to the bottom and rotates in circular motion at the top
- D) Will take infinite time to reach the bottom (speed slows down as it goes down)

12. Let us say we have a linked list (with pointer to only start element) & dynamic array whose size can change at runtime (implemented exactly like vectors). Which of the following operations will take different amortized(average) time complexity to execute? [3 marks]

- A) Deleting an element at the middle of both data structures
- B) Adding an element at the middle of both data structures
- C) Finding an element from the beginning of both data structures
- D) Adding an element from the end of both data structures

13. What all use cases best fit queue data structure?
- a. To keep a track of local variables during runtime
 - b. To transfer data between two asynchronous process
 - c. To handle the hardware or real-time system interrupts
 - d. For evaluation of mathematical expression [3 marks]

A) a,b,c
B) b,c
C) c,d
D) a,d

14. You are developing a recommendation system for an e-commerce platform that suggests products based on user preferences and past interactions. To represent the relationship between users and products efficiently, which graph property would you focus on? [3 marks]

A) The graph should be weighted to represent the preferences between users and products.
B) The graph should have nodes that represent products only, ignoring users.
C) The graph should have many disconnected components for faster processing.
D) The graph should support bidirectional edges to recommend users to products as well.

15. Consider a graph with n vertices where each edge has a weight of 1. If you need to find the shortest path from a source to all other nodes, why would Breadth-First Search (BFS) be preferred over Dijkstra's algorithm (to find the shortest path using priority queue) in this case? [3 marks]

A) BFS has a lower time complexity for graphs with uniform edge weights
B) BFS can handle graphs with both positive & negative edge weights
C) BFS guarantees finding the shortest path in weighted graphs
D) Dijkstra's algorithm will be preferred here rather than BFS because BFS will give incorrect result

16. The function should return the maximum value that can be achieved with a given weight limit. Complete the missing line. [3 marks]

```
def func(weights, values, W):  
    n = len(weights)  
    dp = [[0 for _ in range(W + 1)] for _ in range(n + 1)]  
  
    for i in range(1, n + 1):  
        for w in range(1, W + 1):  
            if weights[i - 1] <= w:  
                _____  
            else:  
                dp[i][w] = dp[i - 1][w]  
  
    return dp[n][W]
```

What should fill the blank?

A) $dp[i][w] = \max(dp[i - 1][w], values[i - 1] + dp[i - 1][w - weights[i - 1]])$
B) $dp[i][w] = \max(dp[i - 1][w], values[i] + dp[i - 1][w - weights[i - 1]])$
C) $dp[i][w] = \max(dp[i - 1][w], values[i] + dp[i - 1][w + weights[i - 1]])$
D) $dp[i][w] = \max(dp[i - 1][w], values[i] + dp[i - 1][w - weights[i]])$